

Auss Abbood

Cambridge, UK — contact@aussabbood.com — www.aussabbood.com

EDUCATION

University of Cambridge, Cambridge, UK
PhD in Computation, Cognition and Language

Oct. 2024 —

Osnabrück University, Osnabrück, Germany
M. Sc. in Cognitive Science
Specialized in artificial intelligence, neuroinformatics, and neuroscience.

Oct. 2016 — May 2019
Mark 1.1, in a scale of 1 – excellent to 5 – bad
With distinction

Goethe University Frankfurt, Frankfurt am Main, Germany
B. Sc. in Biological Sciences
Specialized in neuroscience.

Oct. 2013 — Sep. 2016
Mark 1.6, in a scale of 1 – excellent to 5 – bad

EXPERIENCE

Robert Koch Institute
Data Scientist and Project Lead

Berlin, Germany
Oct. 2018 - Sep. 2024

- Built a risk assessment pipeline for COVID-19 by combining tabular and text data, applying self-developed risk metrics, and predicting short-term risk, saving 1h of manual work daily.
- Co-developed intensivregister.de (ICU surveillance) working on data engineering, visualization, and on-demand data analyses to support ICU management in Germany with up to several millions visits per day.
- Automated key information extraction and relevance scoring for public health surveillance to potentially save epidemiologists 30 minutes of daily work.
- Built collaborations with partners from the public and private sector from more than a dozen countries and with organization such as the European Centre for Disease Prevention and Control, Africa Centres for Disease Control and Prevention, and the World Health Organization Hub.
- Teaching experience including a lecture on outbreak detection, a training of trainers for epidemiologist in Uzbekistan on data science in epidemiology. Supervised two student assistants and co-supervised one Master thesis.

Communintelligence
Co-founder and Data Scientist

Münster, Germany
Oct. 2017 - Jan. 2019

- Data science and deep learning consultation for developers in a public utility company.
- Predicted bus delays with a mean error of only two minutes by analyzing multivariate bus traffic time series using random forests.

PROJECTS

Public Health Intelligence Capacity Building and Innovation
Project Link: PHI-CBI

Berlin, Germany
Jan. 2023 - Oct. 2024

- Leading technical work to increase relevance for users in the text-based surveillance system Epidemic Intelligence from Open Sources (EIOS).
- Conducted user interviews and extensive data analyses to identify bottle-necks during surveillance with EIOS.
- Experimented with recommender systems, anomaly detection, and classification algorithms for text data to identify most promising approach for increase relevance of articles shown in EIOS. Prototyping ongoing.
- Helped acquiring €1,600,000.00 in funding. Supervising two data scientists.

Data and AI-supported early warning system to stabilize the German economy
Project Link: DAKI-FWS

Berlin, Germany
Dec. 2021 - Oct. 2024

- Co-lead for work packages on early warning from epidemiological data such as Germany's notification system, waste water surveillance, and self-reported symptom data.
- Implemented a forecast hub with specialized metrics to evaluate AI models for early-warning on multi-modal data.
- Developed a new post-hoc labeling algorithm for COVID-19 timeseries on which AI-based early warning models can be evaluated.
- Helped acquiring €860,000.00 in funding for our institute. Co-supervised two data scientists.

Natural Language Processing for Event-based Surveillance with Africa CDC
Project Link: NaLaA

Berlin, Germany
May 2021 - Jan. 2024

- Lead project on applying natural language processing on Twitter data for event-based surveillance.
- Developed a language model in Sesotho for geo-tagging of Tweets.
- Worked on an interface for Africa CDC's existing event management system for Twitter surveillance component.
- Co-organized workshop on event-based surveillance for Southern African RCC on surveillance methods.
- Acquired €590,000.00 in funding. Was project lead. Supervised one data scientist.

The Focus Group on Artificial Intelligence for Health by ITU and WHO

Project Link: FG-AI4H

Nov. 2019 - Aug. 2023

- Co-lead Topic Group on Outbreaks and AI test specifications.
- Researched and defined best practices on developing and benchmarking outbreak detection algorithms
- Developed a benchmarking setup with qualitative and quantitative measures for benchmarking outbreak detection algorithms
- Collected best practices to ensure safe use of AI and benchmarking through testing

PUBLICATIONS

- Hicketier A., Bach M., Oedi P., Ullrich A., **Abbood, A.** (2025). Ensemble-labeling of infectious disease time series to evaluate early warning systems. *Infectious Disease Modelling*, 11 (3).
- **Abbood, A.**, Meng, Z., Collier, N. (2025). Time to Revisit Exact Match. *Findings of EMNLP 2025*.
- Bhatia S, [...], **Abbood, A.** [...] et al. (2023). Lessons from COVID-19 for rescalable data collection. *The Lancet Infectious Diseases*, 23 (9).
- **Abbood, A.**, Ullrich A., Denkel, L.A. (2023). Understanding COVID-19 reporting behaviour to support political decision-making: a retrospective cross-sectional study of COVID-19 data reported to WHO. *BMJ Open*, 31 (1).
- **Abbood, A.**, Ghazzi, S. (2023). General Framework for Evaluating Outbreak Prediction, Detection, and Annotation Algorithms. *medRxiv*.
- Oala L, [...], **Abbood, A.** [...] et al. (2021). Machine Learning for Health: Algorithm Auditing & Quality Control. *Journal of Medical Systems*, 45 (2021).
- **Abbood, A.**, Ullrich, A., Busche R., Ghazzi, S. (2020). EventEpi—A natural language processing framework for event-based surveillance. *PLOS Computational Biology*, 16 (11).

CONFERENCES

- **Abbood, A.**, Meng Z., Collier, N. (2025). Time to Revisit Exact Match. – Accepted for EMNLP Findings 2025.
- Zhang M, **Abbood, A.**, Meng Z., Collier, N. (2025). The Power of Simplicity in LLM-Based Event Forecasting. REALM – Workshop at ACL.
- 35th NeurIPS – Online (2021). Workshop.
- EPIDEMICS 8 – Online (2021). Pre-recorded presentation.
- Applied Machine Learning Days – Lausanne (2020). Poster presentation.
- European Scientific Conference on Applied Infectious Disease Epidemiology – Stockholm (2019). Presentation.
- The 14th Annual Conference of the German Society for Epidemiology – Ulm (2019). Presentation.

SKILLS

- **Relevant Coursework:** Algorithms and Data Structures, Foundations of Software Development, Scientific Programming in Python, Machine Learning (ML), Comparative ML, Advanced ML, Implementing ANNs with TensorFlow, Applied Mathematics I&II, Action & Cognition I&II, Advanced Topics in Action & Cognition
- **Programming languages:** Python R, Java, SQL, Bash
- **Software packages:** Altair, BeautifulSoup, gensim, Flair, Git, Luigi, matplotlib, NLTK, numpy, pandas, plotly, PyTorch, SciPy, Selenium, sklearn, spaCy, Statsmodels, TensorFlow
- **Languages:** German (fluent), English (fluent), Arabic (good)
- **Udacity Nano Degrees:** Deep Learning (2019) and Natural Language Processing (2021)

PRIZES AND SCHOLARSHIP

- Winner of the MOOD EU Horizon 2020 summer school hackathon with an ensemble machine learning model on predicting tick prevalence in the UK using spatio-temporal earth observation, vegetation, and weather data (2023)
- Scholarship of the Rosa-Luxemburg-Foundation (2018)
- Second place in the hack4health building a Google Trends-based early warning system and a chatbot for syndromic surveillance for general practitioners (2018)
- Winner of the MÜNSTERHACK with an augmented reality-based bus station information chart and with machine-learning based bus-delay predictions (2017)
- Winner of the Hasso-Plattner-Institute Hackathon with a chatbot on vaccine consultation (2017)